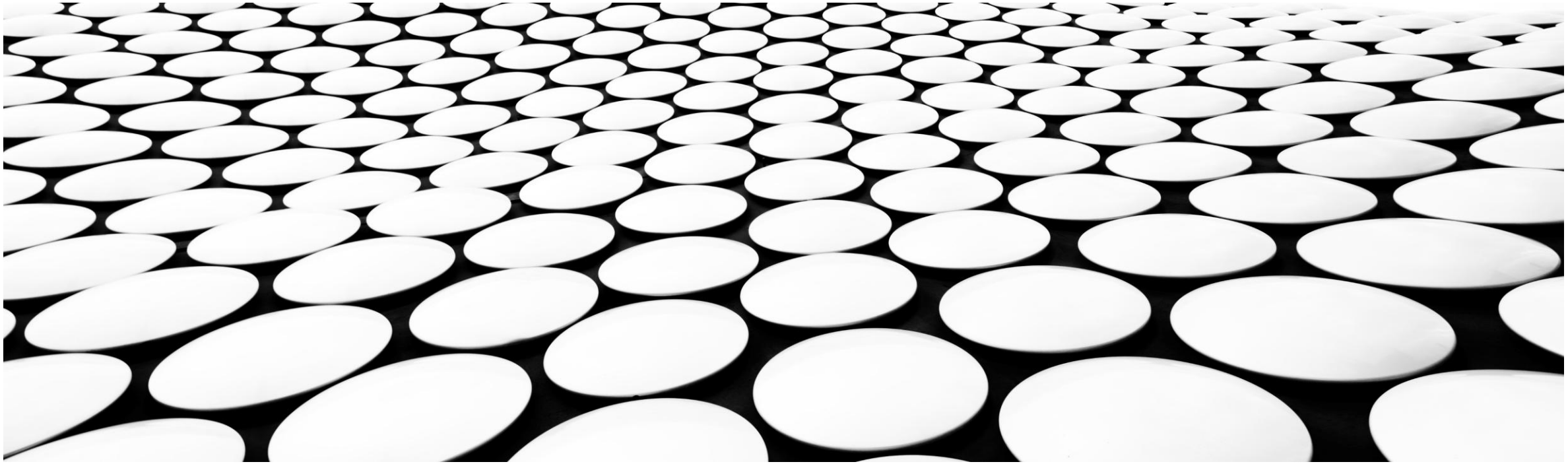
DATA 607 LEADERSHIP IN DATA SCIENCE

FALL 2025 – 09.16.2025



AGENDA

- Questions?
- Next Week – 9/23, you will have an “asynchronous” session (materials will be posted that day for you to respond via discussion post in Blackboard)
- Capabilities of Leading Projects – Part 2
- Breakout Discussion

GROUP PROJECT & INDIVIDUAL LIGHTNING TALK

- Group and individual deliverables
- Document submission
- In class presentations
- Peer Feedback (individual submission)
- Start thinking about the topic for the Individual Lightning Talk on a selected leadership virtue (e.g., How will I practice tenacity in my DS career?)

DATA SCIENCE VS SOFTWARE ENGINEERING PROJECT MANAGEMENT

- **Project team size:** Involves 1–2 data scientists, compared with 3–10 engineers.
- **Project uncertainty:** Data-dependent risks exist on top of engineering risks.
- **Project value:** Demonstrated through A/B tests, feature completion is not enough.

PROJECT MANAGEMENT

WATERFALL

Named "waterfall" due to one-directional project progress flow

Linear sequential phases

Phases: requirements, design, implementation, verification, and maintenance

Top-heavy process

Time allocation: 20-40% for requirement gathering and design, 30-40% for implementation, rest for verification

SCRUM

Agile process

Developed for managing complex knowledge work, especially software development

Emphasis on discovering and adapting to unknown unknowns in projects

Work happens in timeboxed iterations (sprints) of 1-4 weeks

Daily sync in timeboxed meetings (daily scrums) of 10-15 minutes

Many teams choose 2 weeks as sprint length

DS PROJECTS WITH DIFFERENT COMPLEXITIES

Small projects:

- Duration: 1-2 weeks
- Managed within one sprint of the scrum process
- Examples: tracking specification definition, monitoring and rollout
- Fewer uncertainties

Mid-sized projects:

- Duration: 2-4 weeks
- Involves one or two data scientists
- Examples: metrics definition and dashboarding, data insights and deep dives, data enrichment, infrastructure improvements, regulatory items
- Significant communication and coordination with business and technology partners
- Planning helps with execution phases and synchronization
- Project length may extend beyond two sprints
- Shifting requirements and problem definition adaptation based on new learning

Large projects:

- Collaboration of one to three data scientists
- Duration: 4-10 sprints
- Examples: modeling, API development, data consistency
- Technical solutions and project deployments for business impact
- Planning crucial to address key project risks early
- Scrums essential for synchronization among team members (data scientists, product/engineering partners, business stakeholders) throughout the project.

DATA SCIENCE PROJECT EXECUTION TRADE-OFFS

- Speed **vs** quality
- Safety **vs** accountability
- Documentation **vs** progress

BALANCING SPEED AND QUALITY

Tech lead as a manager of projects must understand when to empower business partners for timely decisions and when to focus on craftsmanship.

Granularity of business decisions is crucial in this balancing act, requiring both art and science.

Guiding toward speed:

- Recommendations in data insight and deep-dive projects prioritize top three to five enhancements.
- Emphasis on speed for go or no-go decisions, considering the reach and impact of recommendations.
- Estimates for top recommendations can be refined if they have similar reach and impact.

Guiding toward quality:

- Infrastructure improvement projects focus on tackling tech debt for significant business impact.
- Phase-based approach using modeling PoC and product PoC to remove project risk and improve efficiency.
- Paying down tech debt in a timely manner is essential to enhance overall productivity and reduce operational overhead.

BALANCING SAFETY AND ACCOUNTABILITY

Learning from Failure: The Importance of Post-Mortem Process

- Regardless of careful planning and execution, DS projects can still fail due to unknown risks.
- Learning from failures is crucial for improvement.
- Post-mortem process enables a culture of learning and identifies opportunities for improvement.

BALANCING SAFETY AND ACCOUNTABILITY

The Role of Post-Mortem Process

- **Accurate account:** Understand what happened, observed effects, expectations, and assumptions.
- **Avoid blame:** Blaming individuals is counterproductive.
- **Open communication:** Create a safe environment for data scientists and partners to share the true nature of failure.

BALANCING SAFETY AND ACCOUNTABILITY

- **Empowering accountability:** Provide authority to team members to improve the process or platform involved in the failure.
- **Detailed accounts:** Encourage team members to share their contributions to failures.
- **Expertise and education:** Allow those who made mistakes to become experts and educate the organization to prevent future mistakes.
- Balancing accountability and safety is challenging but essential for growth and learning.

BALANCING DOCUMENTATION AND PROGRESS

Documentation often seen as competing with project progress

Common concerns about the usefulness of documentation:

- Small teams with good communication
- New teams without project handoffs
- Highly technical decisions made within the team
- Limited time for documentation amidst new projects
- Lack of a centralized location for documentation

BALANCING DOCUMENTATION AND PROGRESS

Benefits of Documentation

- **Efficient onboarding:** Growing teams without sufficient documentation can lead to inefficiencies and repeated challenges.
- **Institutionalizing knowledge:** Documentation captures hard-learned knowledge about data, processes, and technical/business decisions.
- **Primary audience:** Future self and new team members joining the project.

BALANCING DOCUMENTATION AND PROGRESS

Properties of Good Documentation

- **Reproducibility:** Documentation enables the ability to reproduce results and processes.
- **Transferability:** Documentation facilitates the transfer of knowledge to new team members.
- **Discoverability:** Documentation should be easily discoverable to locate information effectively.

Characteristics of Good Documentation

- **Length and nuance:** Good documentation doesn't have to be long and nuanced.
- Emphasis on reproducibility, transferability, and discoverability.
- Balancing documentation efforts with project progress is essential for effective knowledge sharing and collaboration.

REPRODUCIBILITY IN DATA SCIENCE DOCUMENTATION

Reproducibility: Obtaining consistent results using the same input data, processing steps, methods, code, and conditions of analysis.

Increases customer confidence in results.

Challenges of reproducibility:

- Changes in input data over time (e.g., appending new data, historical data updates).
- Documenting data observation and aggregation time windows.
- Documentation of modeling methodologies and random seeds.

TRANSFERABILITY IN DATA SCIENCE DOCUMENTATION

Transferability: Applying learnings from one context to another.

Providing readers with project contexts for assessing transferability.

Key components of documentation for transferability:

- Business context and assumptions.
- Validated methodologies.
- Highlighting underlying market mechanisms or fundamental human behavior.

Long-term impact for the organization beyond the specific DS project.

DISCOVERABILITY IN DATA SCIENCE DOCUMENTATION

Discoverability: Documentation appearing in relevant search results for team members.

Various tools available for collaborative documentation.

Highlights and inefficiencies of different tools:

- Wikis: Effective for multi-faceted content, authorship/versioning, and search function. Limited support for tables/graphs.
- Google Docs/Sheets: Authorship/versioning, search function, comprehensive spreadsheet functionality. Limited direct data source integration.
- Scripting notebooks (e.g., Jupyter, Databricks): Documenting data, code, figures, and contexts. Weak search functions, less automated integrations.

BEST PRACTICES FOR DISCOVERABLE DOCUMENTATION

- Main document location: Wikis or Google Docs.
- Record contexts and business decisions.
- Link scripting notebooks, spreadsheets, and production code check-ins to complete the documentation.
- Balancing the strengths and limitations of different tools for effective discoverability.
- Regularly review and update documentation to maintain relevance and searchability.

LEVERAGING EXPERT KNOWLEDGE IN DATA SCIENCE

Expert knowledge: Domain-specific insights gained through years of practicing DS.

Insights include:

- Business opportunities aligned with business goals.
- Data source limitations to be aware of.
- Organizational structure challenges that impact project success.

Role of DS tech lead:

- Guiding technical direction to align with business goals.
- Crystalizing project plan narratives for fast approval.
- Navigating data source limitations and organizational constraints.

Leveraging expert knowledge leads to successful project outcomes.

LEVERAGING EXPERT KNOWLEDGE IN DATA SCIENCE

You can infuse expert knowledge into the projects with a CAN process:

- Clarify the business context of opportunities or risks
- Account for domain data source nuances
- Navigate organizational structures in the industry

CLARIFYING BUSINESS CONTEXT FOR DS PROJECTS

Tech lead's role in aligning technical direction with business context.

Understanding the strategic positioning of DS projects in the organization.

Importance of clarifying the business context:

- Interpreting the vision and mission of the organization.
- Differentiating between vision and mission statements.
- Ensuring consistency with executive team's intended direction.

ALIGNING PROJECTS WITH VISION AND MISSION

Vision and mission as guiding principles for companies.

Challenges in maintaining alignment as companies grow.

Role of DS tech lead in checking alignment:

- Sensitivity to how vision and mission are defined.
- Assessing current projects for consistency.

Characteristics of aligned projects:

- Importance, usefulness, and worthwhileness.

EXAMPLES OF ALIGNMENT WITH VISION AND MISSION

LinkedIn:

- Vision: Create economic opportunity for every member of the global workforce.
- Mission: Connect professionals to make them more productive and successful.
- DS projects aligned with economic opportunities and productivity.

OkCupid:

- Mission: Serve the fundamental human need to find love and happiness through meaningful relationships.
- Example project: A/B testing matching scores.
- Ethical considerations and alignment with the mission.

Tech lead's role:

- Evaluating projects' alignment with the company's vision and mission.
- Applying expert domain knowledge to ensure alignment.

ACCOUNTING FOR DATA SOURCE NUANCES IN DS PROJECTS

Importance of considering domain-specific biases, inaccuracies, and incompleteness in data sources.

Impact of these nuances on project milestones and success.

Role of experienced DS tech lead in mitigating data limitations for project success.

UNDERSTANDING DATA SOURCE NUANCES

Definition of biases, inaccuracies, and incompleteness in data.

Biases: Systematic differences between collected data and represented populations.

Inaccuracies: Misrepresentation of facts in the data.

Incompleteness: Data that is not entirely collected.

EXPERT KNOWLEDGE FOR MITIGATING DATA LIMITATIONS

Expert knowledge as a valuable asset in reducing project failure rates.

Recognizing and anticipating data biases, inaccuracies, and incompleteness.

Mitigation strategies to account for data source nuances:

- Improved data collection methods.
- Data preprocessing techniques.
- Statistical analysis to identify and address biases.

ENSURING PROJECT SUCCESS

Responsibilities of a DS tech lead:

- Proactively addressing data limitations.
- Guiding the team in handling domain-specific data challenges.

Collaborative approach to mitigate data source nuances:

- Engaging domain experts for insights.
- Constant evaluation and refinement of data sources.
- Ongoing monitoring and feedback loop to improve data quality.

CASE 1 WEB SESSION DATA

Definition of web session as a sequence of interactions between client and server.

Fundamental data limitations in web session data: biases, inaccuracies, and incompleteness.

Importance of understanding and managing these data nuances for reliable analysis results.

CASE 1 DATA LIMITATIONS

Biases:

- Robot crawlers can account for a significant portion of website interactions.
- Filtering out robot crawler behavior is necessary to analyze human behavior accurately.

Inaccuracies:

- Bugs in web page implementation can pollute client-side states stored in HTTP cookies.
- Historical client-side bugs may have long-lasting effects on user state analysis.

CASE 1 DATA LIMITATIONS (CONT.)

Incompleteness:

- Missing data due to best-effort delivery mechanism of web traffic.
- Clearing of cookies by users for privacy reasons results in loss of interaction histories.
- Using sequence numbers for tracking data transfer reliability, rather than requiring completeness.
- Impact of handling data nuances on consistency of metrics and modeling results.
- Benefits of being aware of and managing data limitations for producing trusted analysis results.

CASE 2 GEOLOCATION

Geolocation is essential for providing personalized location-based services and understanding user interaction context.

Different approaches, such as GPS, IP addresses, cell tower triangulation, and WiFi hotspots, are used to collect geolocation data.

Nuances in interpreting geolocation data arise from variations in Earth's shape assumptions and obfuscations for privacy or security reasons.

CASE 2 GEOLOCATION (CONT…)

Fundamental limitations in geolocation data include biases, inaccuracies, and incompleteness.

Biases occur when geolocation data is collected only during app usage, leading to a biased subset of locations.

Inaccuracies stem from errors in GPS, IP address modification, cell tower database freshness, and signal availability in WiFi triangulation.

Incompleteness arises from lack of internet connection, delayed transmission of GPS data, and incomplete storage and forwarding of geolocation time series.

CASE 2 GEOLOCATION (CONT…)

Understanding geolocation data nuances is crucial for providing differentiated services and detecting fraud.

DS tech leads can prioritize and plan projects by assessing the riskiness associated with geolocation data limitations.

Consideration of biases, inaccuracies, and incompleteness is essential for drawing reliable conclusions and ensuring the success of geolocation-related projects.

CASE 3 FINANCIAL TRANSACTIONS

Financial wellness apps aim to provide personalized recommendations based on a comprehensive view of a user's financial transactions across multiple institutions.

Data aggregations require user authorization through usernames and passwords.

Variations in data accuracy, completeness, and timeliness can occur due to business relationships and technology integration.

CASE 3 FINANCIAL TRANSACTIONS (CONT…)

Fundamental data limitations in financial transaction data include biases, inaccuracies, and incompleteness.

Biases stem from the partial view of a user's financial situation and variations in data aggregation across numerous financial institutions.

Timing biases arise from differences in aggregation frequency and data delay across the process.

Inaccuracies can occur due to timing discrepancies between pending and settled transactions and user behavior-triggered issues during account reconciliations.

CASE 3 FINANCIAL TRANSACTIONS (CONT…)

Incompleteness of financial transaction data can result from access and connection issues.

Access-related incompleteness occurs when not all user accounts are linked at all times.

Connection-related incompleteness is caused by different levels of integration between data aggregators and financial institutions, leading to missing transaction details.

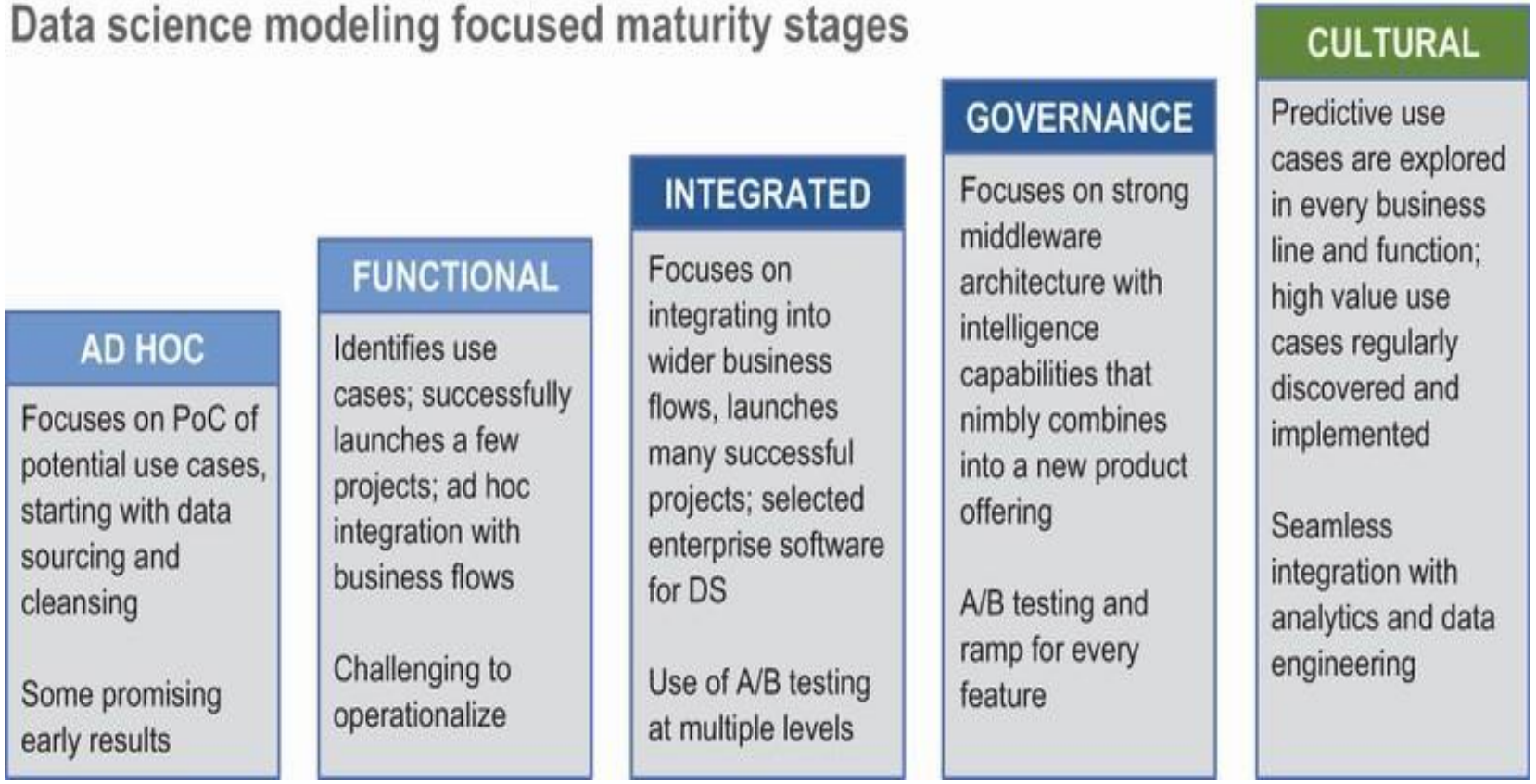
Incompleteness can lead to misinterpretations of a user's cash flow and affect the accuracy of financial recommendations.

SUMMARY

As DS tech leads, you can take care of domain concerns by

- **Recognizing data nuances** as expert knowledge in your domain
- **Respecting teammates with expert knowledge** to fill the team's knowledge gaps
- **Being sensitive to building this expert knowledge** as a source of enterprise value
- **Openly sharing expert knowledge with other DS teams** as well as business partners to better prioritize and plan projects that can provide a more significant business impact

MATURITY OF DS ORGANIZATION



NAVIGATING BUSINESS PARTNER ORGANIZATION STRUCTURE

DS projects in traditional industries can face deployment challenges within existing organizational structures.

Understanding the traditional industries landscape and business opportunities presented by DS is crucial for project success.

Organizational challenges within the industry and alternative paths for success should be considered.

NAVIGATING BUSINESS PARTNER ORGANIZATION STRUCTURE (CONT…)

Understanding the traditional industries landscape is important to align with business partners and address industry pain points.

The business opportunity for DS lies in leveraging smartphone usage data to assess creditworthiness and predict fraud in consumer lending.

Organizational structure challenges arise due to the separation of credit risk and loan operating teams in traditional lending businesses.

NAVIGATING BUSINESS PARTNER ORGANIZATION STRUCTURE (CONT…)

An alternate path for project success involves transferring DS knowledge from operations to the risk functions within a consumer lending company.

Collaboration with fraud investigation teams helps identify likely fraudulent cases and prioritize fraud investigations.

Over 30 weeks of cross-functional deep-dives resulted in the discovery of over 100 highly effective features for fraud prevention, saving the company millions of dollars.

NAVIGATING BUSINESS PARTNER ORGANIZATION STRUCTURE (CONT…)

Organizational expert knowledge is crucial for recognizing partnership and solution deployment risks in DS projects.

Navigating organizational structure challenges is an invaluable capability for a DS tech lead in traditional industries.

Recognizing and addressing these challenges can greatly improve the success of DS projects.

BREAKOUT DISCUSSION

- The case study is in Blackboard under “Course Materials”